

MOHD IMARAN

Ph.D., Centre for Ocean, River, Atmosphere and Land Sciences (CORAL)

Indian Institute of Technology Kharagpur, West Bengal, India

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[Google Scholar Profile](#)

EDUCATION

Ph.D., Ocean, River, Atmosphere and Land Sciences · 2017–2025

Centre for Ocean, River, Atmosphere and Land Sciences (CORAL), Indian Institute of Technology Kharagpur

Thesis: *Numerical Study of Sediment Transport in the Northern Bay of Bengal using Regional Ocean Modelling System*

Supervisor: Prof. Arun Chakraborty & Joint Supervisor: Prof. Subhasish Tripathy

M.Tech., Earth System Science and Technology · 2015–2017

Indian Institute of Technology Kharagpur · CGPA: 7.87

M.Sc., Applied Geology · 2013–2015

Aligarh Muslim University · 71.5%

B.Sc. (Hons.), Geology · 2010–2013

Aligarh Muslim University · 69.40%

PROFESSIONAL EXPERIENCE

Senior Engineer, Geospatial and Geotechnical Intelligence · June 2024 – April 2026

SWANSAT, Swan Corp, India

RESEARCH INTERESTS

- Numerical modeling using the Regional Ocean Modeling System (ROMS)
- HEC-RAS and SWAT hydrological modeling
- Geotechnical and geospatial analysis
- Estuarine modeling: Delft3D, MIKE, Climate Modeling Alliance (CliMA)
- Wave modeling with SWAN and WAVEWATCH III
- Sediment transport modeling under waves and currents; ocean–coast dynamics

TECHNICAL SKILLS

- Programming & Analysis: Python, Julia, MATLAB
- Geospatial Software: ERDAS, ArcGIS, Google Earth Engine, GrADS
- Computing Environments: Windows, UNIX/Linux; high-performance computing (Param-Shakti supercomputer)
- Productivity & Other: Microsoft Office/Excel; applied AI/ML methods

PROFESSIONAL MEMBERSHIPS & LEADERSHIP

- Secretary–Treasurer, Marine and Coastal Geoscience Division, Geological Society of America (October 2025–present)
- Member, Asia Oceania Geosciences Society (AOGS) (2023–present)
- Member, American Geophysical Union (AGU) (2019–2020)

PUBLICATIONS

From Doctoral Research (Ph.D.)

- Imaran, M., Chakraborty, A., & Tripathy, S. (2024a). A high-resolution hydrodynamics and sediment dataset in the coastline waters of the Northern Bay of Bengal. Data in Brief. <https://doi.org/10.1016/j.dib.2024.110577>
- Imaran, M., Chakraborty, A., & Tripathy, S. (2024b). Wavelet analysis of discharge and sediment load for the Subarnarekha, Brahmani, and Mahanadi rivers of India. Hydrological Sciences Journal, 1–25. <https://doi.org/10.1080/02626667.2024.2427355>
- Imaran, M., Chakraborty, A., & Tripathy, S. (2025). Application of ROMS-CSTMS to Analyze Sediment Dynamics and Coastal Morphological Changes in the Northern Bay of Bengal. Modeling Earth Systems and Environment (under revision).
- Imaran, M., Chakraborty, A., & Tripathy, S. (2022). Skill assessment of GloFAS-ERA5 operational river discharge for the major Indian river catchments. International Journal of River Basin Management (under revision).

From SWANSAT Research

- Imaran, M., Chakraborty, A., & Tripathy, S. (2026). An integrated hydro-meteorological and Sentinel-1 SAR detection analysis for near-real-time flood assessment (in preparation).
- Imaran, M., Chakraborty, A., & Tripathy, S. (2026). Irrigation and waterlogging analysis of Sri Ganganagar district, Rajasthan, using Sentinel-1 and Sentinel-2 (in preparation).

CONFERENCE PRESENTATIONS & ACADEMIC PARTICIPATION

- Imaran, M. Participant and Student Volunteer, Asia Oceania Geosciences Society (AOGS), Singapore, 27 July – 1 August 2025.
- Imaran, M., Chakraborty, A., & Tripathy, S. (2023). Study of Cohesive and Non-cohesive Sediment on the Northern Bay of Bengal Shelf Using ROMS-CSTMS Climatological Simulation. Asia Oceania Geosciences Society (AOGS), Singapore, 30 July – 4 August 2023.
- Participant, National Conference on “Challenges in Earth System Sciences for Global Sustainability (CESSGS),” Centre for Ocean, River, Atmosphere and Land Sciences, IIT Kharagpur, 15–16 January 2020.
- Imaran, M., Chakraborty, A., & Tripathy, S. (2019). A Numerical Study on Seabed Erodibility in the Northern Bay of Bengal using ROMS and Community Sediment Transport Modelling System. AGU Fall Meeting 2019.
- Imaran, M., Chakraborty, A., & Tripathy, S. (2019). A Numerical Study on Seabed Erodibility in the Northern Bay of Bengal using ROMS and Community Sediment Transport Modelling System. OSICON-2019 “Indian Ocean Processes and Resources—A Key to Blue Economy”, 12-14 December 2019, CMLRE, Kochi, INDIA.
- Imaran, M., Chakraborty, A., & Tripathy, S. (2019). Participant, research school on “Global Climate Anomalies, Teleconnections & Regional Implications,” TERI-NORCE Climate Research School, New Delhi, India, 5–8 August 2019.
- Imaran, M., Chakraborty, A., & Tripathy, S. (2018). Participant, International Workshop on Biodiversity and Climate Change, IIT Kharagpur, 24–27 February 2018.
- Imaran, M. (2017). Coastal Vulnerability Assessment for Odisha Coast using GIS Platform. International Conference on Remote Sensing and GIS Application in Geosciences, Department of Geology, AMU, Aligarh, 8–10 August 2017.